Junyi (Conny) Zhou

Boston, MA 02215
github.com/JunyiZhou-Conny

Tel: (404) 663-5601
Email: junyizhou@hsph.harvard.edu
linkedin.com/in/junyi-zhou-270208247

EDUCATION

- **Harvard University** — Boston, MA
  M.S. in Health Data Science — *August 2025 – December 2026*
  - Coursework: Deep Learning, Healthcare Machine Learning
- **Emory University, College of Arts and Sciences** — Atlanta, GA
  B.S. in Applied Mathematics and Statistics; Minor in Computer Informatics; GPA: 3.925 — *August 2021 – May 2025*
  - Honors: Dean's List Spring 2022, Spring 2023
  - Coursework: Data Structures, Database Systems, Machine Learning, NLP, Probability and Statistics

TECHNICAL SKILLS

- **Languages**: Python, C++, Java, SQL, JavaScript, R
- **Backend & Cloud**: AWS (Lambda, EC2, S3, SageMaker, Redshift), Docker, Kubernetes, Flask, Git
- **Data Systems**: PostgreSQL, MySQL, MongoDB, Pinecone, Spark, Hadoop, Snowflake, BigQuery
- **ML Systems**: PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, RAG / LLM application pipelines
- **Certifications**: AWS Certified Cloud Practitioner; AWS Certified AI Practitioner; AWS Machine Learning Associate

SELECTED EXPERIENCE

- **Agentic LLM Simulation Platform** — Atlanta, GA
  Full-Stack Engineer & Scrum Master — Emory Pediatric Hospital — *January 2024 – Present*
  - Built an **agentic LLM pipeline** with **RAG** and OpenAI models, focusing on grounded responses, hallucination control, and privacy constraints for multi-turn simulations.
  - Designed **prompting and context-window management** so the system can generate dynamic scenarios and retain relevant conversation state across practice sessions.
  - Shipped a full-stack product (**Python/Flask**, React, MongoDB, AWS) with secure real-time analytics for user performance tracking.
  - Led an Agile team of 6 through code reviews and iterative product feedback loops; deployed demo at CloudFront.
- **Transformer Architecture Reimplementation** — Atlanta, GA
  Machine Learning Researcher — Emory Department of Computer Science — *October 2024 – January 2025*
  - Implemented a Transformer in **PyTorch** from scratch (multi-head self-attention, positional encodings, layer normalization) and validated against `nn.Transformer`.
  - Tuned training hyperparameters and optimized cross-entropy training, reaching a **9.38 BLEU** score on English-to-German translation.
  - Published implementation and experiments: GitHub.
- **High-Throughput ML Inference Tooling on HPC** — Atlanta, GA
  Data Analyst — Bou-Nader Lab, Emory Department of Biochemistry — *February 2025 – August 2025*
  - Built Python pipelines and **CLI tooling** around large model inference (AlphaFold-Multimer / AlphaFold3), including documentation for non-ML users.
  - Automated GPU jobs with **parallel scheduling** on HPC clusters; filtered model outputs with quantitative confidence metrics (pLDDT, PAE).
  - Code and docs: GitHub.
- **Generative Models for Cross-Domain Mapping** — Boston, MA
  Graduate Researcher — Mooney Lab & Alvarez-Melis Lab, Wyss Institute — *February 2026 – Present*
  - Implementing **PyTorch** generative / transport models (VAEs, diffusion, Optimal Transport / CellOT) for out-of-distribution mapping in high-dimensional data.
  - Building scalable preprocessing pipelines over large public and lab datasets; evaluating model variants for subpopulation matching quality.